

Multi-View Entity Resolution

Similarity Aggregation vs. Consensus Clustering (Early vs. Late Fusion)

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Entity Resolution Is a Multi-Field Problem

Entity resolution (ER) identifies records across one or more datasets that refer to the same real-world entity.

Standard three-stage pipeline:

- 1 **Block** — reduce $O(n^2)$ pairs to a candidate set
- 2 **Compare** — compute field similarities for each pair
- 3 **Cluster** — group similar records into entity clusters

The challenge: real records have *multiple* heterogeneous fields. How should information from all fields be combined?

Field	Type
Last name	short text
First name	short text
DOB	date / numeric
Street	medium text
City	categorical
Title	long text
Authors	list
Year	numeric

Reframing ER as Multi-View Clustering

Key insight: each field is one *view* of the underlying entity.

Formal setup. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ be n records with p fields $\{f_1, \dots, f_p\}$.

- Field k : pairwise similarity matrix $S^{(k)} \in [0, 1]^{n \times n}$ over candidate pairs (sparse after blocking).
- **Goal:** recover clustering $\hat{\mathcal{C}}$ matching the true entity clustering $\mathcal{C}^*$.

NA-Aware Combination Principle

If r_i or r_j lacks field k , set $S_{ij}^{(k)} = \text{NA}$. Never impute. Combine only the observed fields for each pair:

$$S(i, j) = \frac{\sum_{k: S_{ij}^{(k)} \neq \text{NA}} w_k S_{ij}^{(k)}}{\sum_{k: S_{ij}^{(k)} \neq \text{NA}} w_k}$$

Entity Resolution Foundations

- Fellegi & Sunter (1969) — probabilistic model; independent-field assumption
- Jaro (1989); Winkler (1990) — Jaro / Jaro–Winkler string similarity
- Christen (2012) — block–compare–classify; weighted field combination

Similarity Aggregation (Paradigm A)

- Lanckriet et al. (2004) — MKL: $S = \sum_k w_k S^{(k)}$; kernel sum is a valid kernel
- Gönen & Alpaydın (2011) — MKL survey: supervised, unsupervised, localized

Gap: No prior work systematically compares similarity aggregation and consensus clustering as dual ER paradigms across field types, scales, and missingness regimes.

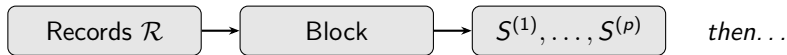
Consensus / Ensemble Clustering (Paradigm B)

- Strehl & Ghosh (2002) — CSPA / MCLA / HGPA; co-association matrix
- Fred & Jain (2005) — Evidence Accumulation Clustering (EAC)
- Topchy et al. (2005) — partition combination via mixture model

Multi-View Clustering

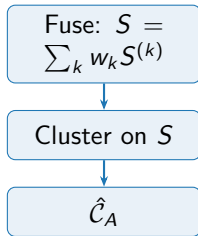
- Bickel & Scheffer (2004) — multi-view EM clustering
- Kumar et al. (2011) — co-regularized multi-view spectral clustering
- Xu et al. (2013) — survey: early / late fusion taxonomy

Two Paradigms for Multi-View ER



Paradigm A — Similarity Aggregation

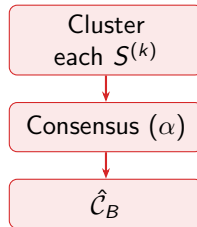
(early fusion)



Combines field similarities *before* clustering.
Single unified matrix; one clustering step.

Paradigm B — Consensus Clustering

(late fusion)



Clusters each field *independently*, then merges partitions by majority vote.

Paradigm A: Similarity Aggregation (Early Fusion)

$$\underbrace{S^{(1)}, \dots, S^{(p)}}_{\text{per-field}} \xrightarrow{\text{weighted sum}} S = \sum_{k=1}^p w_k S^{(k)} \xrightarrow{\text{cluster}} \hat{C}_A$$

Weight learning (`er_weights()`):

Method	When to use
Equal ($w_k = 1/p$)	No prior knowledge
Fellegi–Sunter EM	Unsupervised; no ground truth
ARI-based	Supervised; ground truth on training fold
Bimodality / variance	Proxy for field discriminability

Clustering (8 methods): HC-Average (primary), HC-Ward (comparison), PAM, threshold CC, Louvain, Leiden, label propagation, graph coloring.

Prior work: generalizes Fellegi–Sunter independence model; related to Multiple Kernel Learning (MKL).

Paradigm B: Consensus Clustering (Late Fusion)

$$S^{(k)} \xrightarrow{\text{cluster}} \hat{C}^{(k)}, \quad k = 1, \dots, p \xrightarrow{\text{consensus}} \hat{C}_B$$

Consensus step — majority-vote co-membership with threshold α :

$$r_i \sim r_j \text{ in } \hat{C}_B \iff \frac{1}{p} \sum_{k=1}^p \mathbf{1}[r_i \sim_{\hat{C}^{(k)}} r_j] \geq \alpha$$

Advantages

- Independent per-field clustering $\Rightarrow$ parallelizable
- One bad field only dilutes, not dominates
- No weight learning needed
- α is interpretable

Limitations

- Loses inter-field correlations
- Each $S^{(k)}$ must produce a useful clustering alone
- Consensus can fragment if fields conflict

Prior work: Strehl & Ghosh (2002) CSPA/MCLA; Fred & Jain (2005) evidence accumulation.

Research Questions

RQ1 — Accuracy

Does Paradigm A or B recover entities more accurately across datasets and clustering methods?

RQ3 — Scalability

How does each paradigm scale with n ? Does blocking-induced sparsity and parallel field clustering favor Paradigm B at large scale?

RQ2 — Field Quality

How do field discriminability and missingness affect each paradigm? Which degrades more gracefully?

RQ4 — Predictability

Can field-level diagnostics (cardinality, missingness, per-field ARI) predict which paradigm wins *before* running experiments?

Benchmark Datasets

All four datasets have multiple fields — the multi-view structure is genuine.

Dataset	Records	Entities	Fields	Mode	Role
CORA	1,879	182	15	Dedup	Multi-field ablation
Restaurant	864	754	5	Linkage	Clean small benchmark
DBLP-ACM	4,910	2,690	4	Linkage	Medium bibliographic
NCVR	100,000	98,521	4	Linkage	Large-scale stress test

CORA — title, booktitle, authors, address, date, year, editor, journal, volume, pages, publisher, institution, type, tech, note (15 fields). Rich field set; ideal for ablation (RQ2).

Restaurant — name, address, city, phone, type. Fodors vs. Zagats; clean and small.

DBLP-ACM — title, authors, venue, year. Moderate size; structured text.

NCVR — givenname, surname, suburb, postcode (4 fields). 100K records (10 parties $\times$ 10K); primary scalability benchmark (RQ3).

Experiment Design

Exp 1 — Main accuracy comparison (all 4 datasets, 5-fold CV)

Factor	Levels
Paradigm	A (similarity aggregation), B (consensus clustering)
Weight / α	Equal, Fellegi–Sunter EM; $\alpha \in \{0.3, 0.5, 0.7\}$
Clustering	Threshold CC, Louvain, HC-Average (primary), HC-Ward (comparison)
Metric	ARI, B-cubed F_1 (primary); V-measure, Pairwise F_1 (secondary)

HC-Average preferred: ER clusters are heavily size-skewed and the similarity matrix is non-Euclidean — both violate Ward's equal-size-minimization assumption. Average linkage imposes no such constraint and is standard in the record linkage literature. HC-Ward is retained to quantify the impact of linkage choice.

Exp 2 — Field ablation (CORA): remove one field at a time; compare ARI drop under A vs. B.

Exp 3 — Scalability (NCVR, $n \in \{1K, 5K, 10K, 20K, 50K, 100K\}$): wall-clock time and peak memory vs. n .

Exp 4 — Missingness robustness (DBLP-ACM): mask fields at 0%–50%; measure ARI of A and B as missingness grows.

Working Hypotheses

Paradigm A wins when...

- Fields are *complementary*: each alone is weak but together they discriminate
- Field similarities are on comparable scales
- Ground truth is available for weight learning
- n is moderate

Paradigm B wins when...

- One or a few fields are *individually sufficient*
- Fields are noisy or incommensurable
- No ground truth is available
- n is large (parallelizable)
- Missingness is high

Central claim

The better paradigm can be predicted from field-level diagnostics (cardinality, missingness rate, per-field ARI) before running experiments.

Implementation: erbot R Package

Both paradigms already implemented; paper provides the first systematic comparison.

Stage	Paradigm A	Paradigm B
Similarity	er_similarity() — per-field $S^{(k)}$ vectors	
Combination	er_combine() er_weights()	+ (skip)
Clustering	er_cluster() on S	er_cluster() on each $S^{(k)}$
Aggregation	(done)	er_consensus(α)
Compare	er_compare_paradigms() (in progress)	

Datasets via er_load():

- "cora", "restaurant" ✓, "dblp_acm" ✓,
ncvr_read()

Ground truth via er_load_gold():

- "cora", "restaurant", "dblp_acm"
- NCVR: embedded in partition CSVs

Paper Structure and Target Venue

Proposed sections:

- 1 Introduction
- 2 Related Work
- 3 Multi-View ER Framework
- 4 Algorithms (A, B, weight learning, blocking)
- 5 Experiments (Exp 1–4)
- 6 Discussion: when does each paradigm win?
- 7 Conclusion

Target: IEEE TKDE or VLDB Journal

Companion: R Journal software paper
(`erbot.tex`, existing)

Progress:

Task	Done?
Framework design	✓
Both paradigms coded	✓
Restaurant data	✓
DBLP-ACM data	✓
<code>er_compare_paradigms()</code>	○
Exp 1 (main table)	○
Exp 2 (ablation)	○
Exp 3 (scaling)	○
Exp 4 (missingness)	○
Write paper	○

Contribution

First systematic empirical comparison of *similarity aggregation* and *consensus clustering* for multi-field ER, across four benchmarks at small to large scale.

Paradigm A — Similarity Aggregation

- Fuse $S^{(k)} \rightarrow$ cluster on S
- 5 weight strategies, 8 clustering methods
- Better when fields are complementary

Paradigm B — Consensus Clustering

- Cluster each $S^{(k)} \rightarrow$ consensus $\hat{C}_B$
- Tunable α ; parallelizable; no weight learning
- Better when fields are redundant or n is large

Next: implement `er_compare_paradigms()` and run Exp 1 on all four datasets.

After this paper: theoretical conditions under which one paradigm dominates the other.